

OROMIA STATE UNIVERSITY
COLLEGE OF ST
DEPARTMENT OF IT

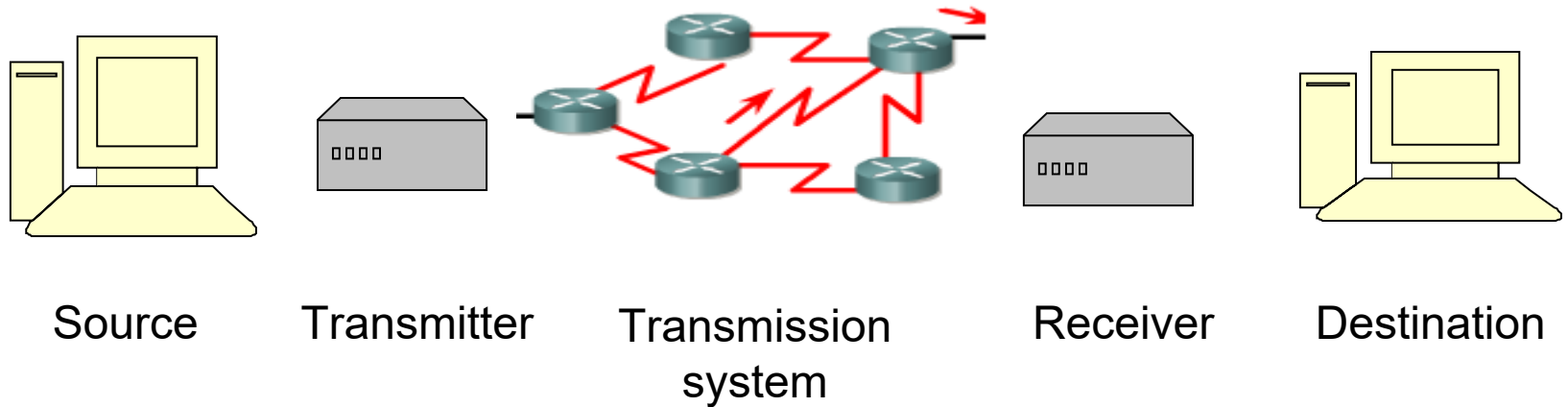
FUNDAMENTAL CONCEPTS OF DATA
COMMUNICATIONS AND COMPUTER
NETWORKS

All topic summary Lecture Note

Introduction

- What is Data *communication*?
- Communication model (e.g. Human communication)
- Source
 - ▣ Generates data
- Transmitter
 - ▣ Converts data into transmitting signals
- Transmission system
 - ▣ Carries data
- Receiver
 - ▣ Converts received signals into data
- Destination
 - ▣ Takes incoming data

Communication model in networks



Data Representation

- Text
 - ▣ Represented in bits patterns e.g. 0,1
 - ▣ Different Bit patterns called code.
 - ▣ Present Coding system: Unicode, 32 bits
- Numbers
 - ▣ Represented in bit patterns
 - ▣ Converted into binary for calculations
- Images
 - ▣ Represented into matrix of pixels/bits
- Audio/Video
 - ▣ Continuous data

Flow of Data

- Simplex
 - One way traffic only, one device transmits and one receives e.g. Keyboard->monitor
- Half-duplex
 - Both stations can transmit and receive but one at time. e.g. Bus topology
 - Only one path from source to destination.
 - collisions may occur
- Full-duplex
 - Both can receive and send at the same time. e.g. Star topology.
 - Two separate transmission lines.
 - collisions free

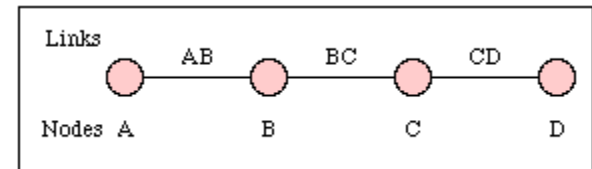


Networks

- Nodes interconnected together and share information and resources.
- Types of Network
 - ▣ Point to point connections
 - ▣ Circuit switching network
 - ▣ Message switching network
 - ▣ Packet switching network

Point to point connections

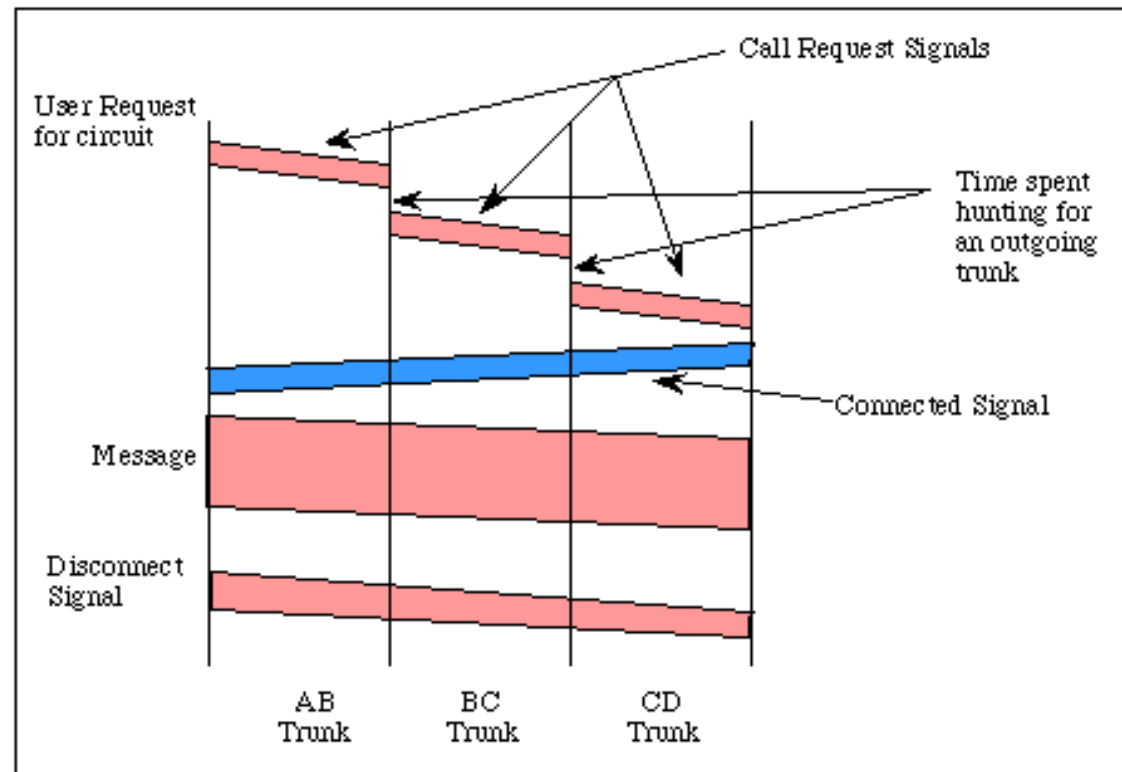
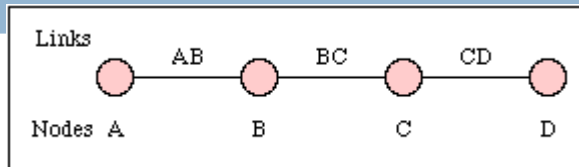
- Not peer to peer
- Dedicated communication circuit
- Fixed configuration
- Direct link between devices
- B and C can be intermediate device to connect A and D
- Connection formed in different sections between users, end to end connection in series and forms circuit.
- So point to point forms simple connection
- If number of users increased then hard to provide circuit that connects each user with other users.
- So we need switching which could provide sharing of transmission circuits.



Circuit switching network

- This allows the communication circuits to be shared among users.
- E.g. Telephone exchange
- Switching
- It allows equipments and circuits to be shared among users.
- Establishes dedicated circuit between users before communication.
- When circuit is free other users can use this. e.g. telephone calls.
- Telephone exchange is an example of circuit switching.
- Replacement conference calls

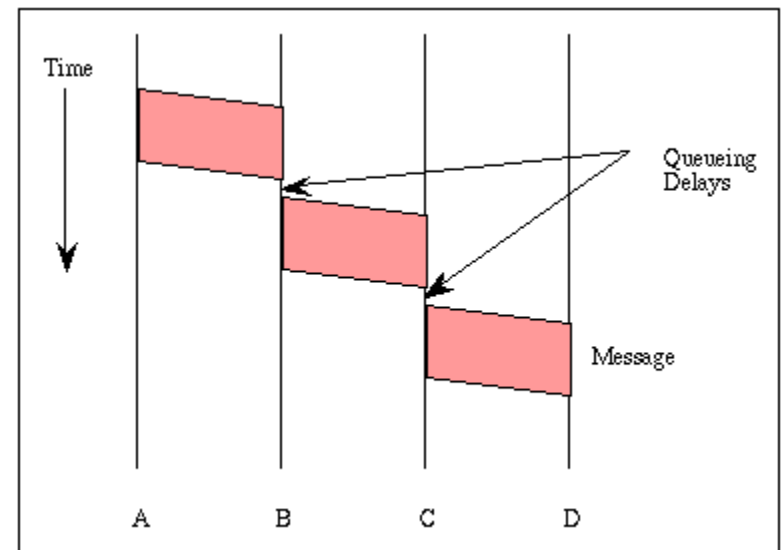
Circuit switching network



- Source connects with switching node
- User requests circuit
- Node B receives connection request and identifies path to node D via intermediate node C.

Message switching network

- Circuit setup, store and forward e.g. Telex or email
- Also called stored and forward switching
- Not necessary to establish circuit between A and D.
- When circuit is free it delivers otherwise waits and store message.
- But delays may occur.



Packet switching network

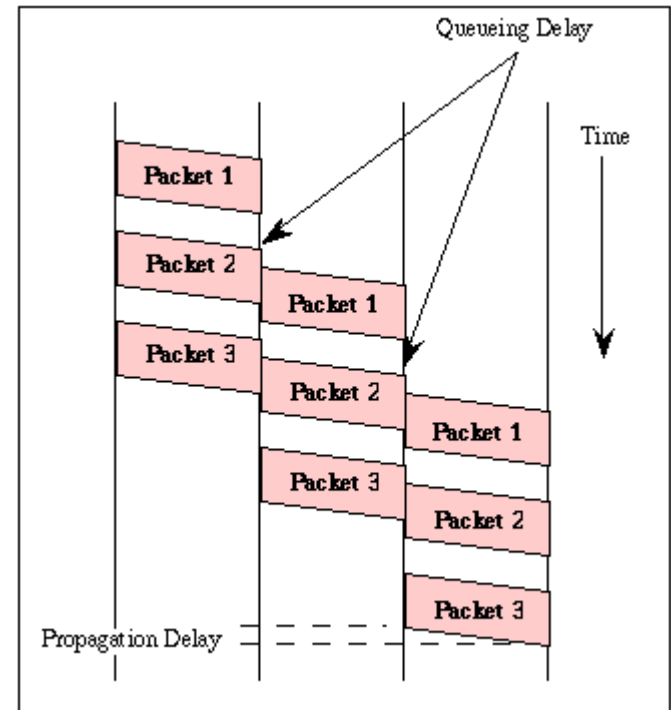
- Similar to message switching
- but divides message into packets/datagram packets of equal lengths.
- Headers are added to each packets.
- Header contains information about source and destination.
- No need for dedicated circuit.
- As length of packet is small so each link is established for small time and then it is available for other messages.
- Another benefit is pipelining.

Packet switching network

Pipelining:

When data sent from B to C at the same time data packet is being sent from A to B.

This results in gain of efficiency. And total delay for the transmission of Message is very less.



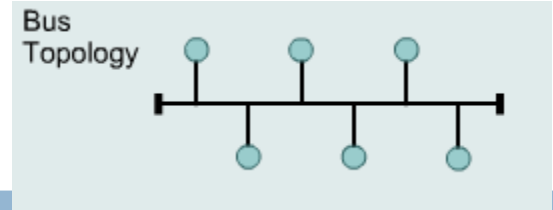
Types of Packet switching network

- LANs
- WANs
- ISPs
- (will be discussed in detail once we set strong base for these networks)

Physical Topologies

- Difference between Network topology and physical topology.
- Network Topology: Defines structure of network
- Physical topology: Layout of the wire or media.
- But physical topology is a part of network topology.
- Physical topology:
 - ▣ BUS
 - ▣ Star
 - ▣ Ring
 - ▣ Mesh
 - ▣ Tree

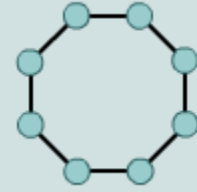
BUS



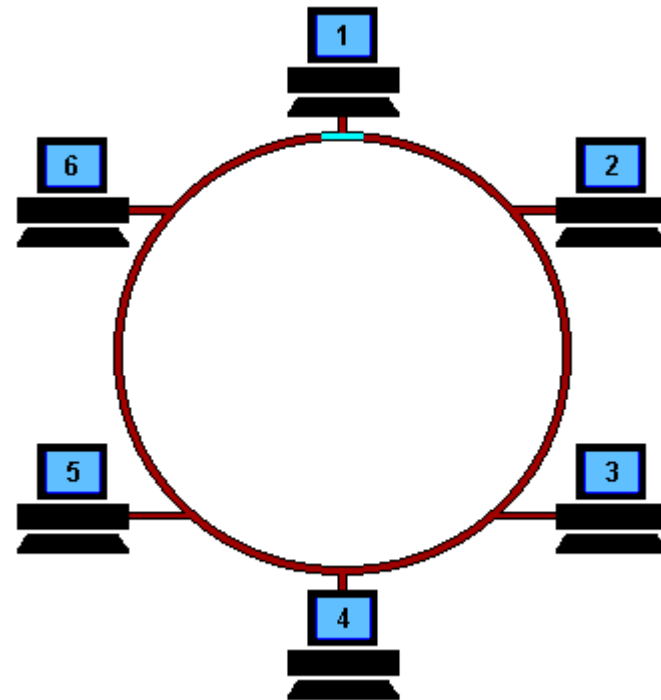
- Uses single backbone cable, All hosts directly connected to this backbone.
- Inexpensive and easy to install
- All nodes receives data
- Ends terminated with a device terminator.
- Two types of BUS
 - ▣ Linear
 - All nodes connected to common medium which has only two end points.
 - ▣ Distributed
 - All nodes connected to common medium which has more then two end points.

RING

Ring Topology

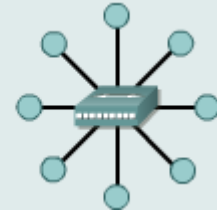


- All nodes connected to one another in form of closed loop.
- Expensive and difficult to install but offers high bandwidth, not robust.
- Point to point connection with only two devices.
- Signal is passed in one direction only, moves until it reaches to its destination.
- Each device connected with a repeater.
- One signal always circulates for fault detection. If device don't receives signal for specified time it generates alarm.



STAR

Star Topology



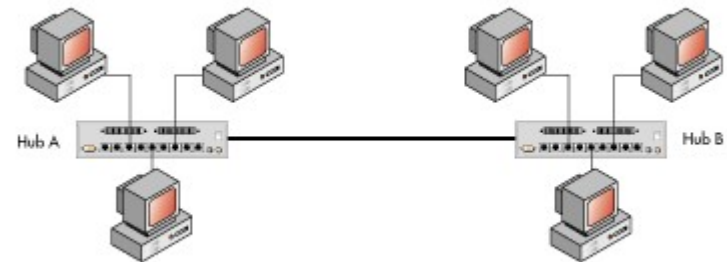
- Connects all devices with central point.
- Central point can be hub.
- Data transmitted reaches to central point, who decides where to send data.
- Bottleneck occur because all data pass from hub.
- Less expensive and easy to install, robust if one link is down still remains active.
- Disadvantage: dependency one centre
- Star is used in LANs



Extended Star Topology



- Types of STAR Topology
- Extended STAR
 - ▣ Has one or more repeaters from central node to extend maximum transmission distance.
 - ▣ If repeaters in extended star topology is replaced with hub or switches then it creates *Hybrid topology*.
 - ▣ Or if backbone as star topology and extended with bus then it also creates *Hybrid topology*.
 - ▣ Connecting two or more topologies with each other forms hybrid topology.
- Distributed STAR
 - ▣ Individual networks based on star topology
 - ▣ These networks do not have central or top level connection points.



MESH

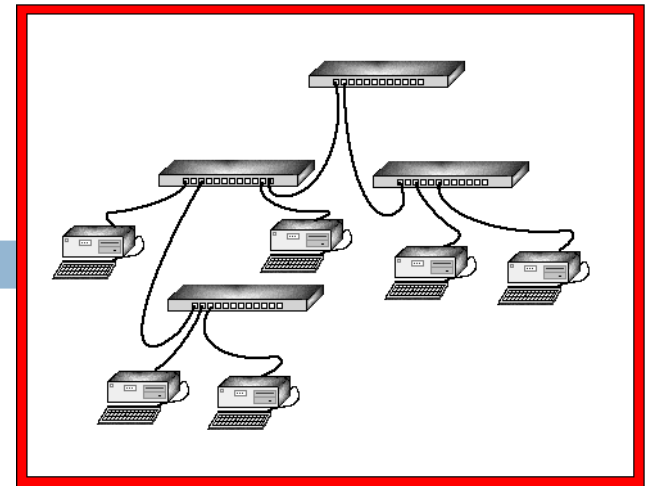
Mesh Topology



- Each host has its dedicated point to point link with every other host.
- Link only carries data between *two devices only (no other can use that link)*
- If there are n number of nodes in network then we need $n(n-1)$ links.
- If link is multi directional or duplex mode then we need $n(n-1)/2$ links.
- Each device requires $n-1$ I/O ports to be connected to each device.
- Eliminates traffic problem, Robust, privacy/security of message.
- More cabling required, more I/O ports needed, hard to install, expensive.

TREE

- Central node connected to one or more nodes one level lower in hierarchy.
- Combines characteristics of linear bus and star topology.
- Must have three levels of hierarchy.
- If only two levels then it forms star.
- If branching factor one then linear hierarchy.
- Physical hierarchy will be one less than total number of nodes in network.
- Disadvantage: requires point to point wiring, requires more hardware, dependent on backbone, difficult to configure.



OSI Layer model



- OSI: Open System interconnection
- Comprises of seven layers
- For network communication all network devices must speak same language or protocol.
- Each layer defines how data is treated and goes through different stages while traveling in network from one place to another.
- All layers are like set of instruction of assembly.
- Gives complete picture of information flows within network.
- All layer are used in end to end systems but only first three layers used in intermediate systems while network communication.

□ OSI layers are divided into two different sets.

□ Application Set

□ Application set consist of Layer 5,6 and 7.

□ Transport set

□ Consist of layer 1,2,3 and 4



Layer 1: Physical Layer

- Physical Layer
 - ▣ Define physical characteristics of network. E.g. wires, connector, voltages, data rates, Asynchronous, Synchronous Transmission
 - ▣ Handles bit stream or binary transmission
 - ▣ Used to maintain, activate and deactivate physical link.
 - ▣ For receiver it reassembles bits and send to upper layer for frames.
 - ▣ For Sender it convert frames into bit stream and send on transmission medium.



Binary Transmission

→ • Wires, connectors, voltages, data rates

Properties Physical Layers

- Deals with *bit stream*.
- Transmits raw bit stream over physical cable
- defines cables, cards, and physical aspects
- defines NIC attachments to hardware, how cable is attached to NIC
- defines techniques to transfer bit stream to cable
- Layer 1 Device: Repeater, Hub, Multiplexer

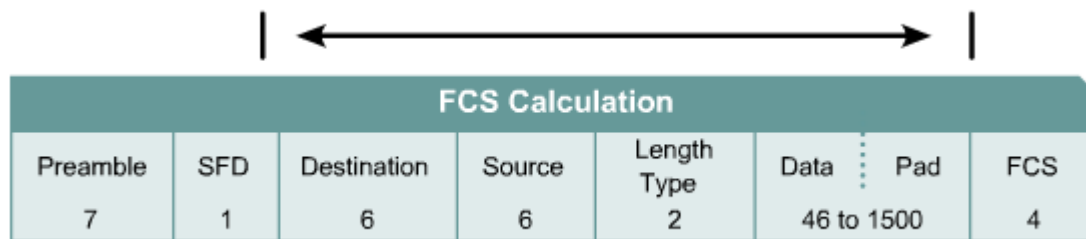
Layer 2: Data Link Layer

- Maintaining, activating, deactivating data links connection.
- Used to transfer data between two entities.
- Used for error handling (CRC), media access control, flow control.
- MAC headers and trailers are added
- Two major operations:
 - ▣ Concerned with physical components
 - ▣ Communicate with upper layers
- Turns packets into bit stream at sending station
- Turns *bits into Frames* for upper layers at receiving layer.
- Layer 2 devices: Bridges, Switches, intelligent hubs, NIC

Layer 2 Frames

Frames include information about:

- Which computers are in communication with each other
- When communication between individual computers begins and when it ends
- Which errors occurred while the computers communicated (LLC)



Sub layers of Layer 2

- Logical link layer (LLC)
 - ▣ Used for communication with upper layers
 - ▣ Error correction
 - ▣ Flow control
- Media Access Control (MAC)
 - ▣ Access to physical medium
 - ▣ Header and trailer

Difference between Layer 1 and Layer 2

- Layer 1 cannot communicate with upper layers
- Layer 2 does this using LLC
- Layer 1 cannot identify computer
- Layer 2 uses addressing process
- Layer 1 can only describe stream of bits
- Layer 2 uses framing to organize bits

Layer 3 Network Layer

- Defines network logical address (not MAC)
- Provide switching and routing facilities
- Determines network address and best path to deliver packets
- Translate logical address into physical address
- This layer responsible for:
 - ▣ Addressing
 - ▣ Route selection
- If router cannot send data in same size as sent by source then layer 3 divides data into smaller sizes, at receiving end network layer reassembles data.
- Forms Packets
- Protocols that operates at layer 3:
 - ▣ IP, ARP, RARP, ICMP,
- Layer 3 Devices:
 - ▣ Routers, ATM switches,

Layer 3 Packets

Packet contains following information:

- Source (source IP address)
- Destination (Destination IP address)
- Length (length of packet)
- Number (Total number of packets in message)
- Sequence (sequence number of packet)

Layer 4 Transport

- Used for data transfer between end systems.
- Processes to processes delivery (not source to destination delivery)
- Provides QoS
- Whole message is received in order.
- Converts data into segments.
- Ensures data is delivered error free and in order.
- Flow control: send that amount of data which can be handled by destination. Similarly if data packet lost then resend.
- Protocols at layer 4: TCP, ARP, RARP, UDP
- Layer 4 Network component: Gateways

Layer 5 Session Layer

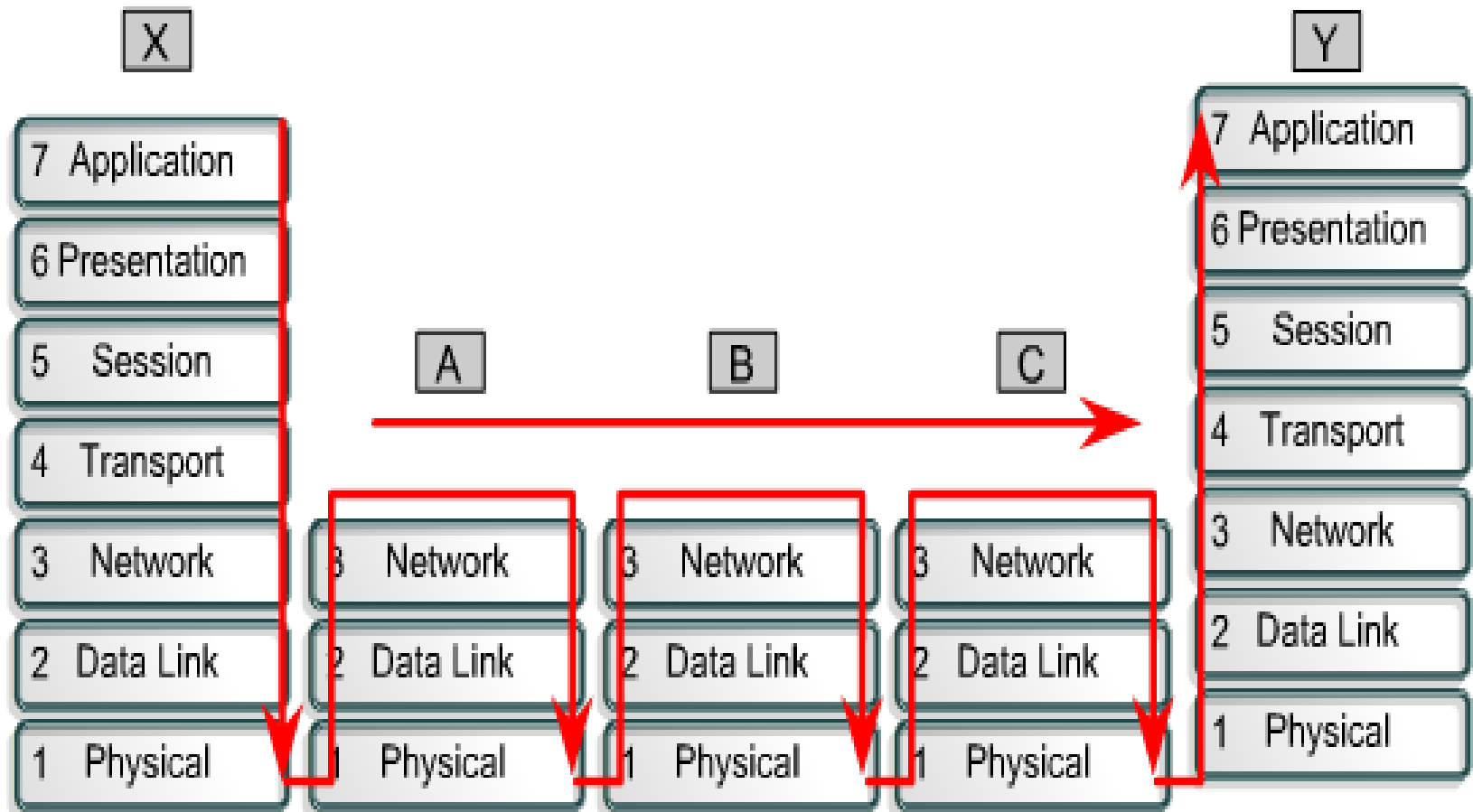
- Used for dialogue control and synchronization purposes.
- Establishes sessions between systems.
- Dialog control:
 - ▣ Dialog between two parties for communication to take place in either half or full duplex mode.
- Synchronization:
 - ▣ Add synchronization points to stream of data.
 - ▣ If session fails only send that data which was not delivered not whole message.
 - ▣ E.g. files of 2000MB

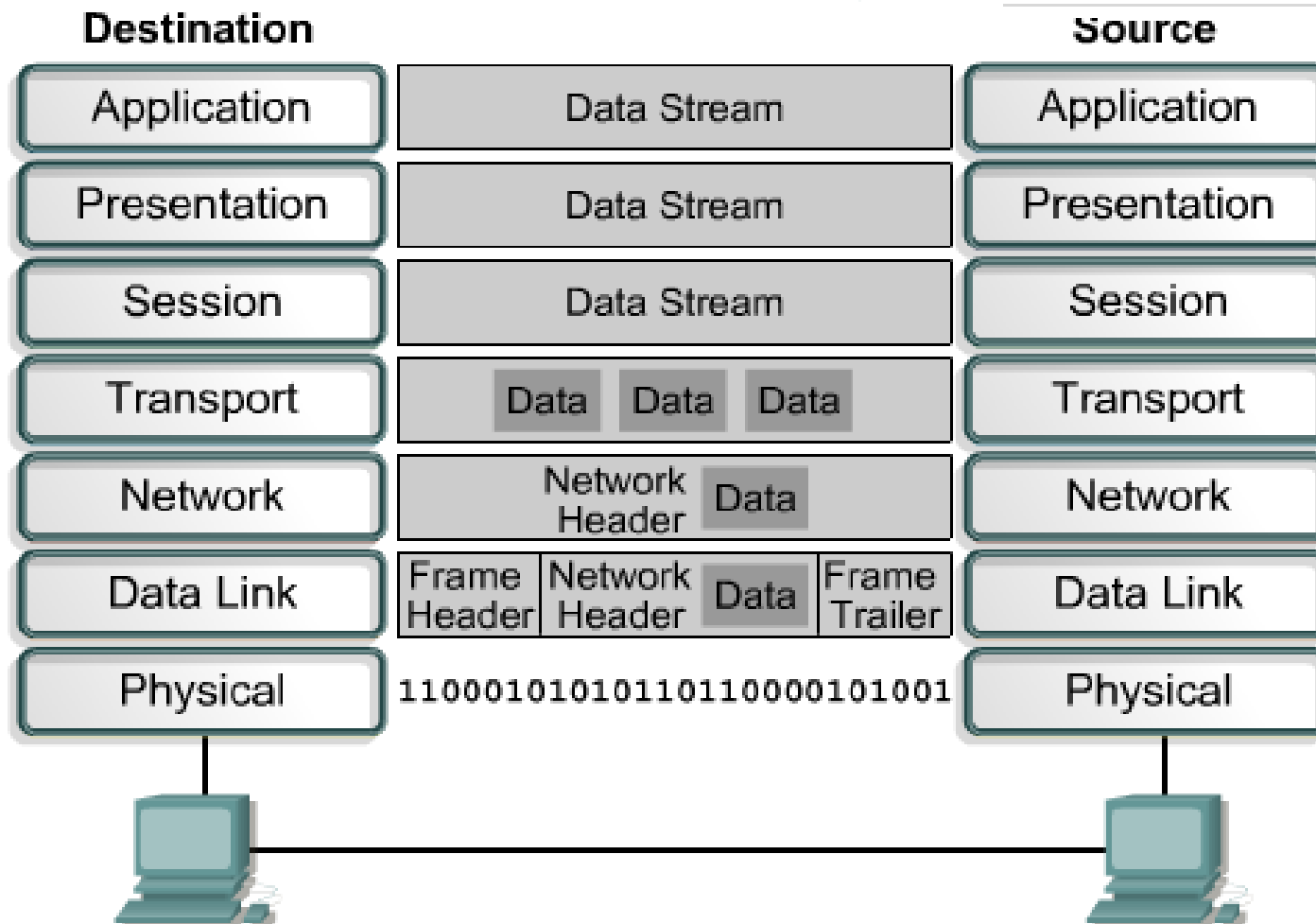
Layer 6 Presentation Layer

- Concerned with syntax and semantics of information.
- Responsible for translation (data into bits and encoding format), compression, and encryption.
- Translation: data into bits and selecting appropriate encoding technique and changing from sender format to receiver format.
- Compression: Reduce number of bits.

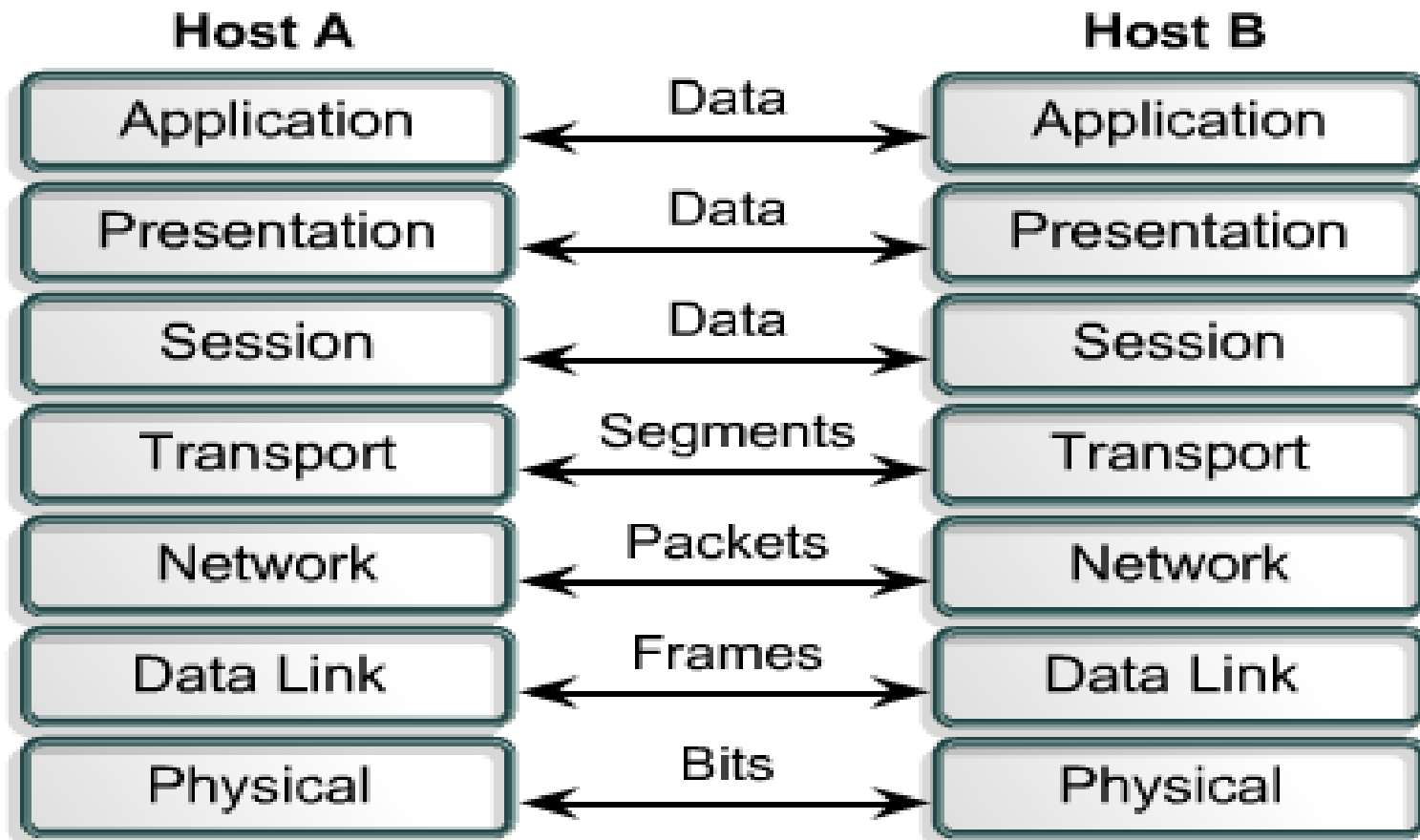
Layer 7 Application Layer

- Layer support Software applications to access network.
- Examples: Virtual terminal (Remote desktop), FTP,TFTP, email (SMTP), Directory services, TELNET.





Transformation of Data in OSI layers



Advantages of OSI



- Network communication is broken into smaller, more manageable parts.
- Allows different types of network hardware and software to communicate with each other.
- All layers are independent and changes does not affect other layers.
- Easier to understand network communication.

TCP/IP

Transmission control protocol:

- Guarantees end to end delivery of data segments
- Arrange segments in order.
- Used to check transmission errors.
- Connection oriented (same route, in order) doesn't mean circuit.
- Reliable process to process communication service.
- Made reliable through sequence number and acknowledgement

Internet Protocol (IP)

- Data sent over internet from source to destination.
- IP is connection less (packets independent, different routes, out of order).

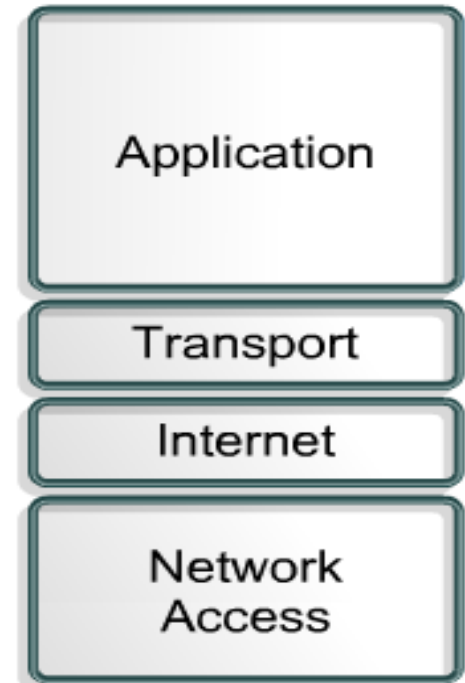
TCP/IP Layers

- Application layer of TCP/IP includes functionality of session and presentation layer of OSI model. Like encoding, dialog control.

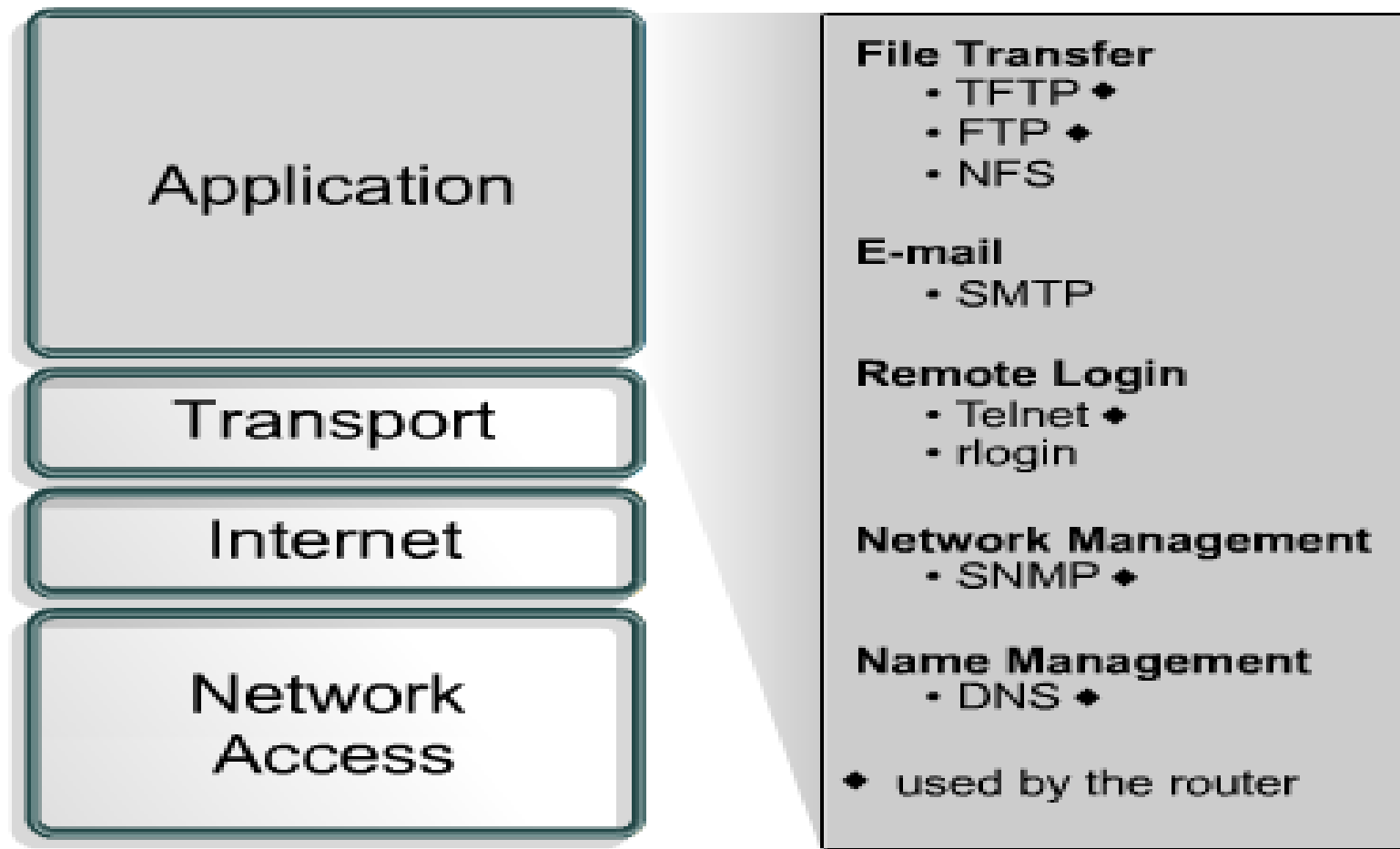
Application layer includes

file transfer, email, remote login, network Management, name management

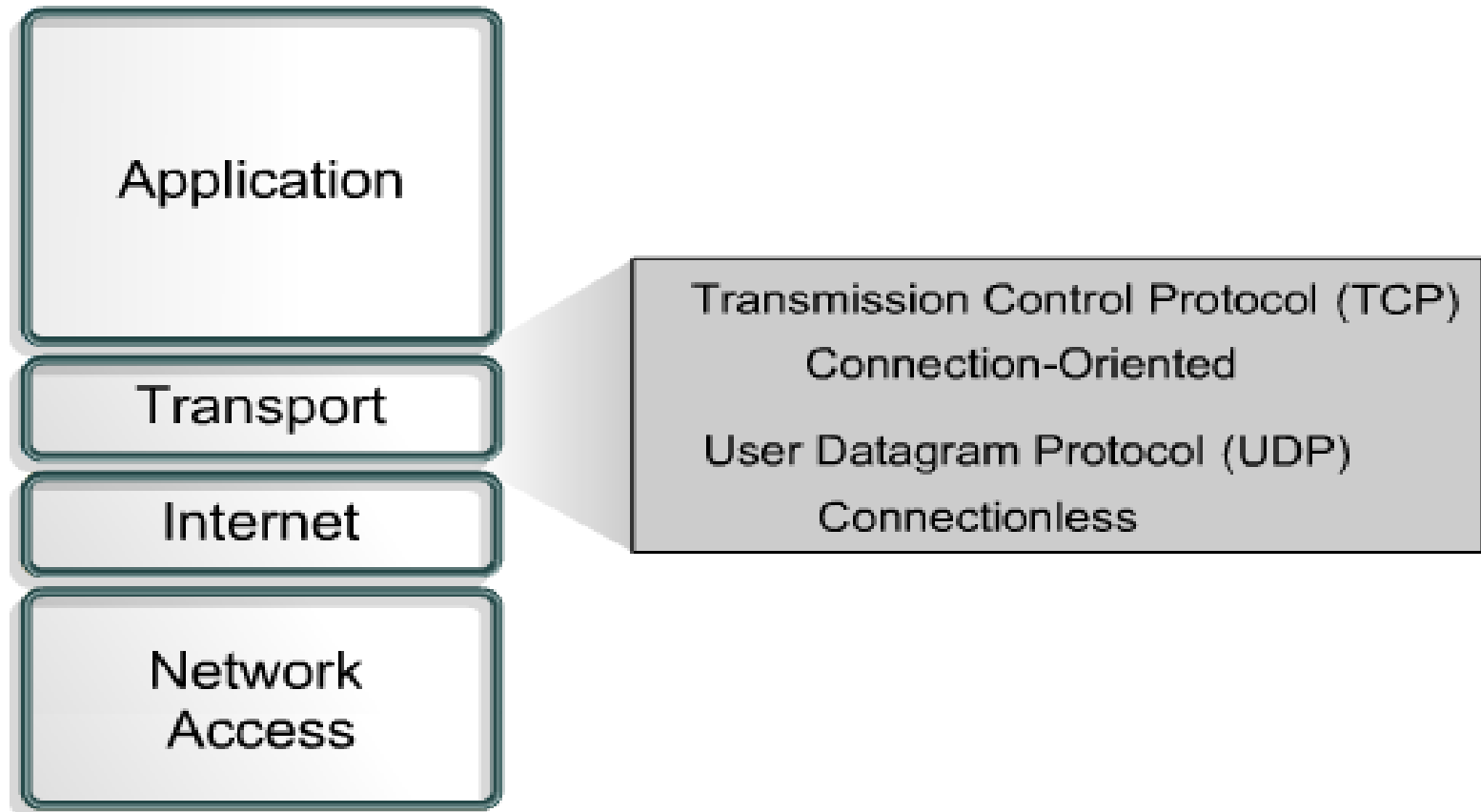
- Transport layer includes QoS, Flow control Processes to processes communication
- IP layer includes ARP, RARP, ICMP
- Network layer physical link to media.



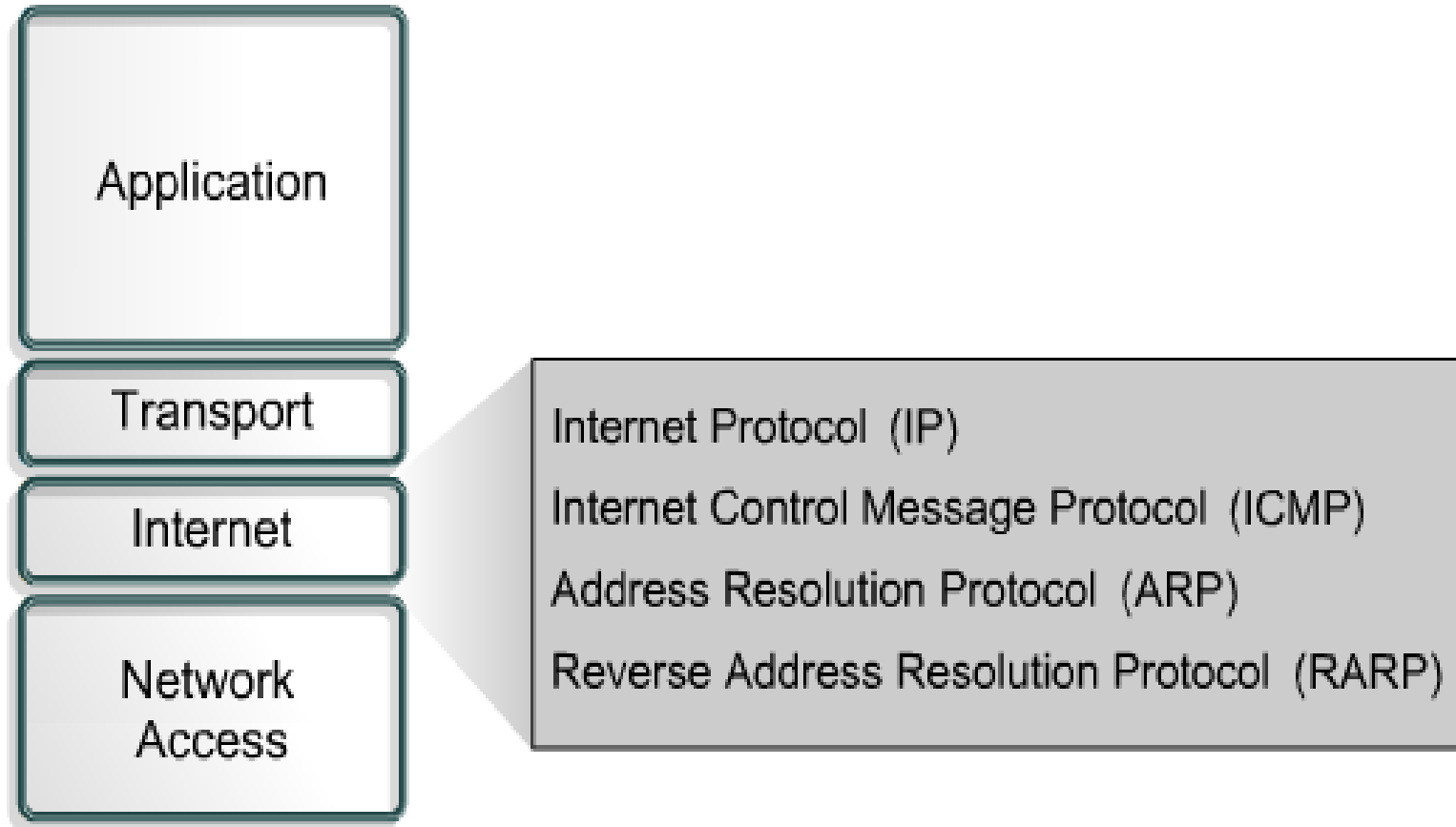
Application layer protocols with their function



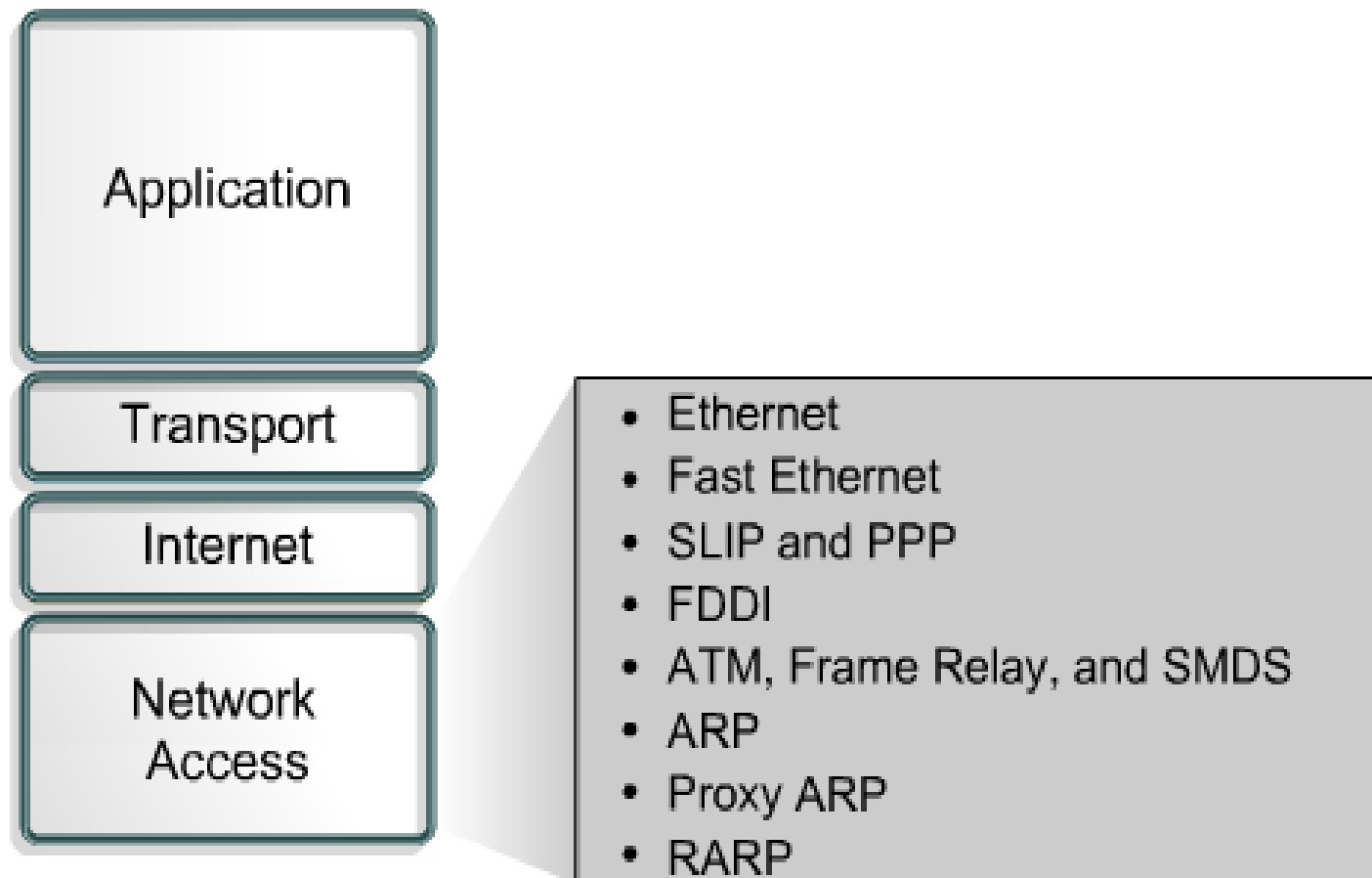
Transport layer protocols with their function



Internet layer with their protocols



Network Access layer with their protocols



OSI Vs TCP/IP

Similarities include:

- Both have layers.
- Both have application layers, though they include very different services.
- Both have comparable transport and network layers.
- Both assume packets are switched. This means that individual packets may take different paths to reach the same destination. This is contrasted with circuit-switched networks where all the packets take the same path.

OSI Vs TCP/IP..

Differences include:

- TCP/IP combines the presentation and session layer issues into its application layer.
- TCP/IP combines the OSI data link and physical layers into the network access layer.
- TCP/IP appears simpler because it has fewer layers.
- TCP/IP protocols are the standards around which the Internet developed, so the TCP/IP model gains credibility just because of its protocols. In contrast, networks are not usually built on the OSI protocol, even though the OSI model is used as a guide.

Layered Protocols

- Internet Protocol (IP) (Layer 3 protocol)
 - ▣ Used for data communication in packet switched network
 - ▣ Unreliable and connectionless (no specific path)
 - ▣ Unreliable
 - Data corruption
 - Packet lost
 - Out of order
 - ▣ Packet called Datagram
 - ▣ internetworking computers
 - ▣ IPv4, IPv6

IPv4

- Internet protocol version 4
- Uses 32 bit address.
- Possible addresses $2^{32} = 4,294,967,296$ (4.3 billion)
- Some addresses are reserved like private addresses plus multicast addresses.
- Private addresses (LANs)
 - 10.0.0.0 – 10.255.255.255
 - 172.16.0.0 – 172.31.255.255
 - 192.168.0.0 – 192.168.255.255
 - Total reserved private addresses = 18 Million
- Multicast addresses
 - 224.0.0.0 – 239.255.255.255
 - Total multicast addresses = 270 million
- Available addresses = possible addresses – (private addresses + multicast addresses)

IPv6

- Increase in number of addresses
- 128 bits long address
- Possible addresses 2^{128}
- 2^{96} more address than IPv4
- ARP, RARP, IGMP are deleted or merged into ICMPv6 protocol.
- Example : 207. 142. 131. 235. 207. 142. 131. 235. 207. 142. 131. 235. 207. 142. 131. 235

ARP Protocol (layer 3)

- Stands for address resolution protocol
- Finding physical address from logical address
- Host or router transmit IP datagram packet containing logical address obtained from DNS.
- Query is broadcast but reply is unicast.
- Request contains **sender and receiver IP plus sender physical address.**
- Reply contains physical address.
- Proxy ARP. (router sends its physical address)



- ARP is used in four cases of two hosts communicating:
 - ▣ When two hosts are on the same network and one desires to send a packet to the other. (same network)
 - ▣ When two hosts are on different networks and must use a gateway/router to reach the other host (internet)
 - ▣ When a router needs to forward a packet for one host through another router. (internet)
 - ▣ When a router needs to forward a packet from one host to the destination host on the same network. (internet)
- **Reverse of ARP**
- Finding logical address from physical address
- Request broadcast to network.
- Based on Client server protocol.

ICMP (Layer 3)

- Used to report errors with delivery of IP data.
 - E.g. if particular service or host not reachable or to check routers are correctly routing .
- **Ping tool** uses ICMP to check host is reachable and how long it takes to reach.
- ICMP message is delivered in IP packet.
- Error reporting not error correction.
- **Two types of messages**
 - ▣ Error reporting message
 - Problems with router or host e.g. destination unreachable, time exceeded, parameters problem
 - ▣ Query message
 - Help in getting specific information. e.g. neighbors

ICMP Errors

□ Network Errors:

- Host or network unreachable

- Network congestion message:

- When router buffers too many packets, and don't process with same speed as received, generates source quench message. Too many messages results congestion.

- Time exceed

- ICMP timeout message is generated when host is unreachable.
 - If errors in routing table, packets travel in loop. At each router value is decremented by 1.
 - When TTL value reaches to 0, packet discarded with ICMP error.
 - TTL value is default

IGMP Layer 3

- Internet group management protocol
 - Protocol involved in multicasting.
 - Protocol that manages group membership.
 - Provides information to multicast routers about the membership status of hosts.
 - Router receives thousand of multicast packets, if destination unreachable broadcast packets. Increases traffic load.
- IGMP help router in providing this information.
- Agent maintains, edit membership and provide information of group.

IGMP (contd...)

- IGMP has following messages
 - ▣ Query
 - Request for information of hosts
 - ▣ Joining report
 - If one process in group sends membership report.
 - ▣ Leaving report
 - When no other processes in company

BOOTP and DHCP

- BOOTP
- Acquire IP automatically
- It enables diskless workstations to
 - Discover its IP address
 - Discover IP of BOOTP server
 - Load file into memory for booting
- DHCP
- Clients obtain following automatically
 - ▣ IP address
 - ▣ Default gateway
 - ▣ Subnet mask
 - ▣ IP address of DNS server



- DHCP address allocation
- Manual allocation
 - ▣ Table is configured at server with MAC addresses manually
- Automatic allocation
 - ▣ Permanently assigns IP from free IP addresses range
- Dynamic allocation
 - ▣ Dynamic reuse IP addresses using TCP/IP software configured at client.

TCP Layer 4

- Transmission control protocol
- Used for exchange of data with applications.
- Reorders data
- Divides data into segments of equal sizes.
- Applications send octets to TCP for transmission, TCP divides into equal segments.
- TCP keeps check that if bytes are damaged, through checksum.
- Sender and receiver both check damaged bytes.

TCP Packet fields

- Source: 16 bit
- Destination: 16 bit
- Sequence number: 32 bit
- Acknowledgement number: 32 bit, receiver increment by 1 as acknowledge.
- Header: 20-60 bytes
- Reserved: 6 bits
- Control: 6 different bits

UDP

Minimum overhead.

- Used to send short messages.
- Not reliable as TCP (out of order, missing datagram, , duplicate datagram).
- Lack of flow control and error control
- Faster and efficient
- Communication takes place using ports.
- Header contains following information:
 - ▣ Source port number (16 bits)
 - ▣ Destination port number (16 bits)
 - ▣ Total length(16 bits)
 - ▣ checksum(16 bits)
- Pseudo header contains rest of information about source address, destination address, etc

DNS Layer 7

- Domain name system
- Stores information about hosts
- Maps names of hosts into IP addresses.
- E.g. google.com is the name space,
- Domain name can have tree like structure.
- Resolver sends DNS request to DNS server.
- Domain should be unique, but duplication among domains is possible.
- Resolver request sent to server, if cannot resolve then referred to another server.
- Mail.google.com: level three domain
- There can only be 127 levels each level can have 63 characters
- Lists are maintained by the registrars.
- Mainly domain name has two parts
 - ▣ Rights most represent top level domain
 - ▣ Left specifies subdomain
- Every domain has one or more domain name server

Modes of transmission

- Unicast
 - ▣ Information sent from one sender to one receiver
 - ▣ Use standard unicast applications e.g. ftp, http, smtp and telnet
- Broadcast
 - ▣ Information sent from one sender and all other connected receiver
 - ▣ ARP uses broadcast to resolve address
 - ▣ 255.255.255.255
- Multicast
 - ▣ Information sent from one or more sender to a particular set of users.
 - ▣ E.g. video server transmitting TV channels

Transmission Impairment

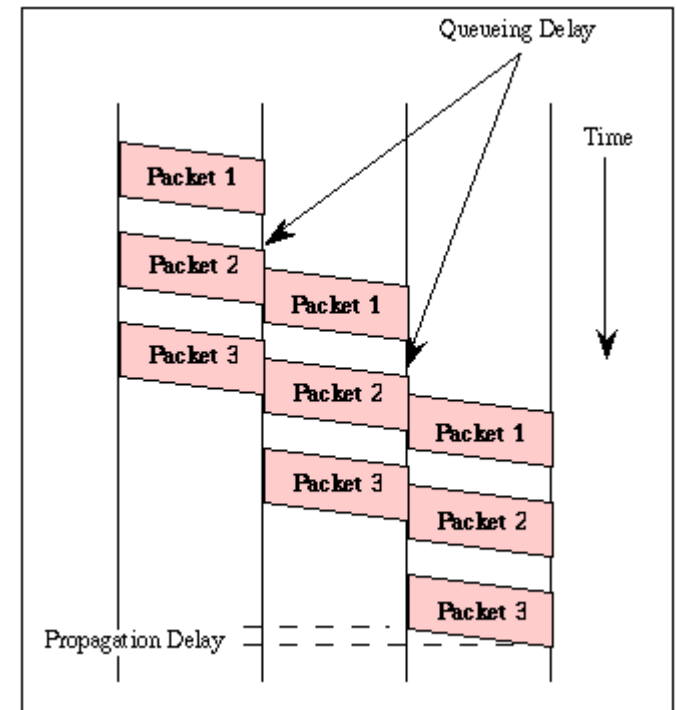
- Attenuation
- Propagation delay
- Distortion
- Noise
- Crosstalk
- Jitter

Attenuation

- Reduction in strength of signals
- Also referred as Loss
- Signals traveling on long distance loses their strength.
- Signals losses some of their energy and signals are converted into heat.
- Represented in Decibels
- Cables measured in 'decibels per foot'.
- More efficient cable = less attenuation per unit distance.
- Repeaters are used to overcome attenuation.
- Repeaters regenerates signals.

Propagation delay

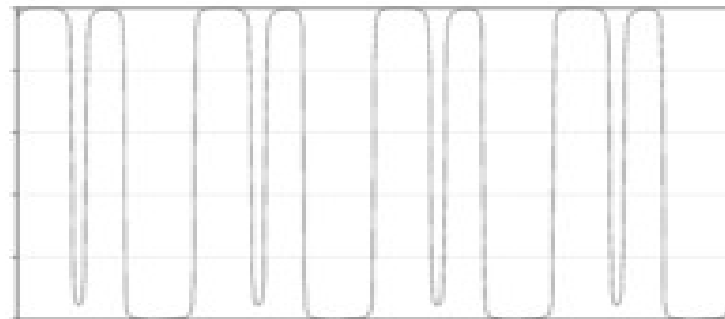
- Delay from the time signal transmitted and the time signal received.
- Measured in milliseconds.
- Varies from medium to medium
- Distortion
- Change in shape of signal



Distortion



Original waveform



Distorted waveform

Noise

- Addition of external factors in signals
- Noise can disturb data.
- Two wires can generate voltage noise which affects data.
- Noise which corrupts data can be:
 - ▣ Thermal noise (signals generated by electrons by random motion)
 - ▣ Induced noise (generated by motors and appliances)
 - ▣ Crosstalk (affect of one wire on another)
 - ▣ Impulse noise (generated by power lines)

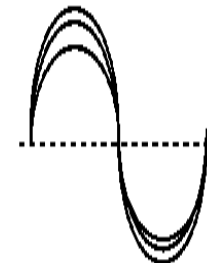
Crosstalk & Jitter

Crosstalk

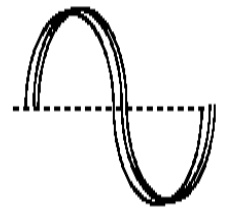
- One line induces signal into another
- Mostly happens in pair cables.

Jitter

- Variation in the signals or data packets at destination with variation of time. E.g. application at destination is time sensitive like audio or video stream.
- Jitter can be of two types
 - ▣ Amplitude jitter
 - Small constant change in amplitude, can be caused by power noise
 - ▣ Phase jitter
 - Small constant change in phase of signal,



Amplitude Jitter



Phase Jitter

Performance

- Bandwidth
 - Bandwidth in hertz
 - Range of frequencies contained in signal
 - Bandwidth in bits per second
 - Number of bits per second a channel or network can transmit
- Throughput
 - How fast a data can be sent through a network
 - Bandwidth and throughput are different
 - Link with bandwidth 1Mbps but device can only process 200 Kbps.
- Latency
 - Delay between the message transmitted and message received.
- Latency can be caused due to:
 - Propagation time
 - Transmission time
 - Queuing time
 - Processing time.



□ Propagation time

- Time required by bit to travel from source to destination
- That is total distance per unit speed

□ Transmission time

- Time required to send complete message
- Measured in message size per unit bandwidth available

□ Queuing time

- Time required by intermediate device to process data.
- varies with load on network.
- E.g. packets queuing

Transmission media

- Two types of media

- ▣ Guided

- Uses cabling system to guide data signals to a specific path.

- ▣ Unguided

- Data signals travels not to a specific path.

- Types of Guided media

- ▣ Open wire

- ▣ Twisted pair

- ▣ Coaxial cable

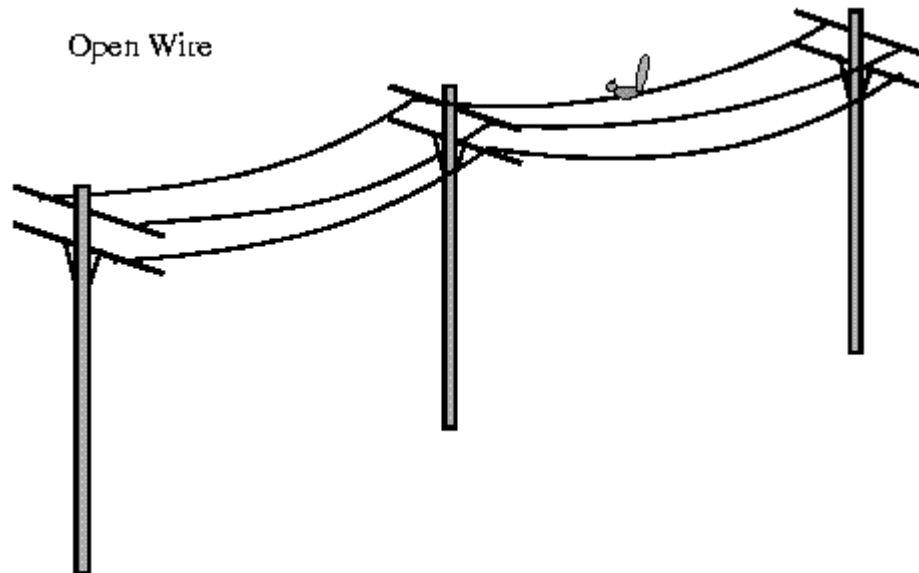
- ▣ Optic fiber



- Important consideration related to cables performance:
 - ▣ Speed for data transmission
 - ▣ Digital (Baseband) or analog transmission
 - ▣ How far signal travels before it gets attenuated.
- Specification related to cable type are:
 - ▣ 10BASE-T
 - ▣ 10BASE5
 - ▣ 10BASE2

Open wire

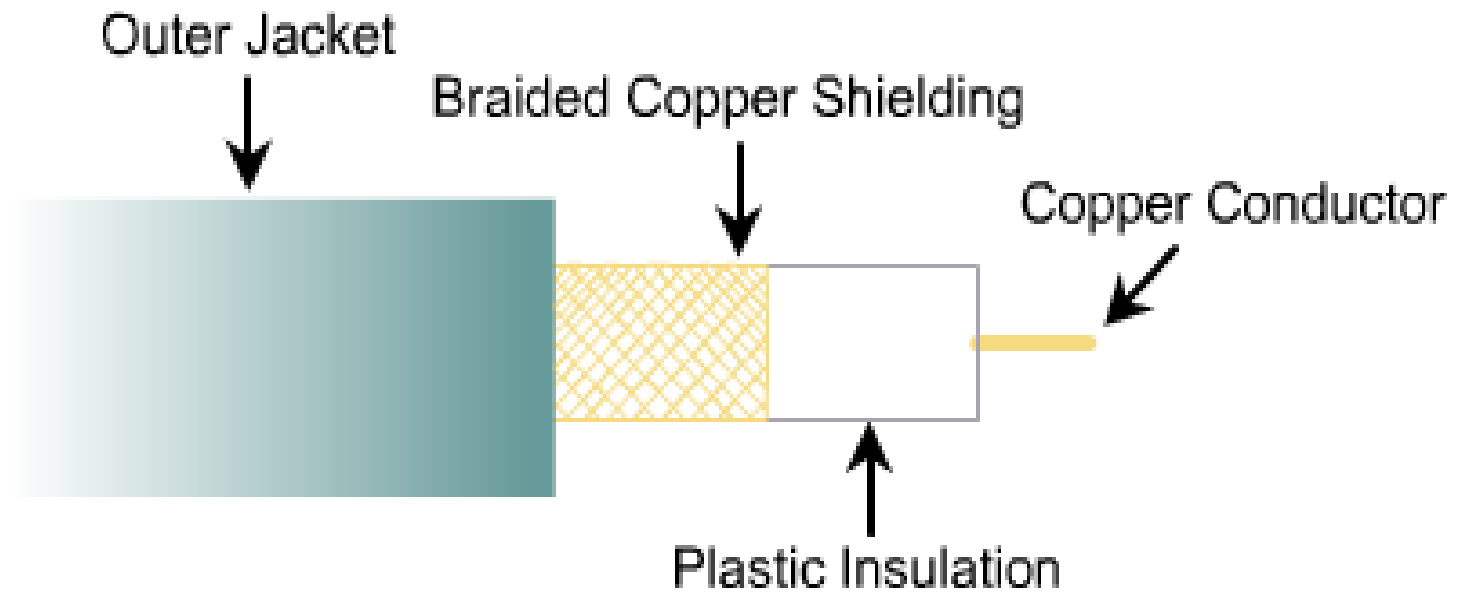
- Open electric wires
- No shielding or protection from external noise
- Cannot be used for data transmission but for less distances.



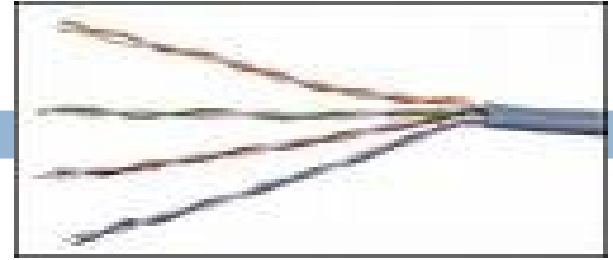
Coaxial cable

- Outer shield protects inner shield from outer electric signals.
- Similarly insulator between two conductors protects them from noise generated by either conductor.
- Cable has 10 – 100 Mbps speed
- Inexpensive
- Maximum cable length 500m.
- Coaxial cable offers several advantages for LAN.
- Run longer distance than other cables.

...



Twisted pair



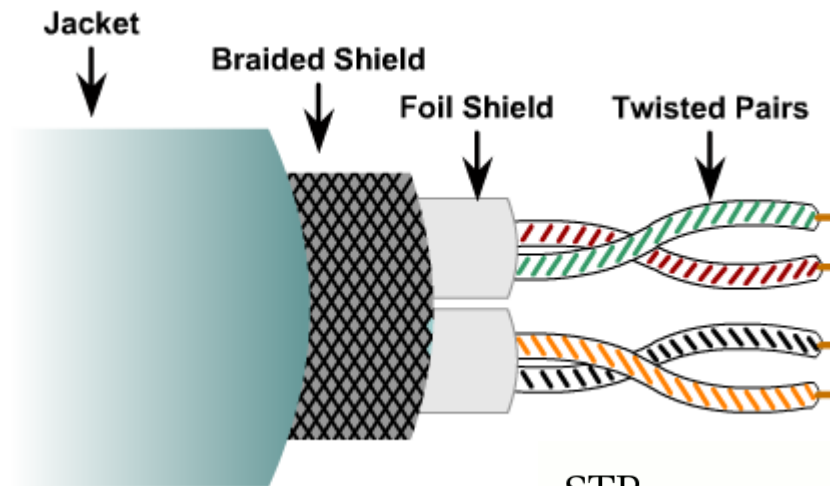
- Wires are twisted in pairs
- Each pair carries +ve and -ve signals
- Noise appearing on one wire will also occur on other wire of same pair.
- Noise appeared on both wires of pair will cancel its affect.
- Twists of pair cancels the noise affect.
- Increase in the number of turns per foot reduces noise interference.



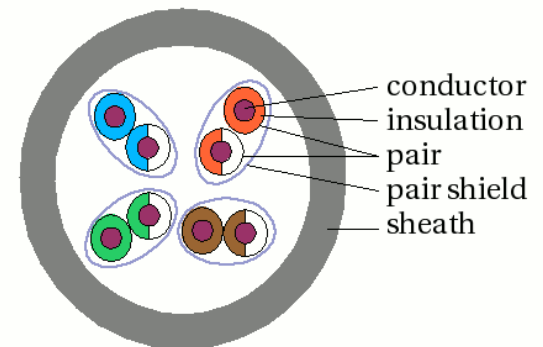
Types of Twisted pair

- Shielded twisted pair
 - ▣ STP cable combine the techniques of cancellation, shielding and twisted wires.
 - ▣ Each pair wrapped in metallic foil, then two pairs are wrapped in overall metallic foil.
 - ▣ STP reduces
 - Electric noise within pairs and outside noise
 - crosstalk
 - ▣ STP provide protection from all kind of noises
 - ▣ It is expensive and hard to install.
- 0 – 100 Mbps Speed
- Maximum cable length 100m before signals attenuated.

Shielded twisted pair



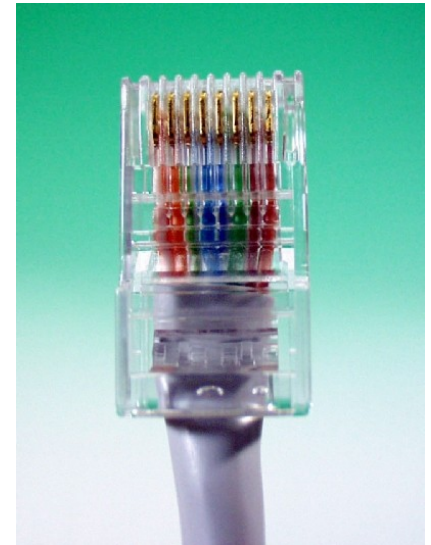
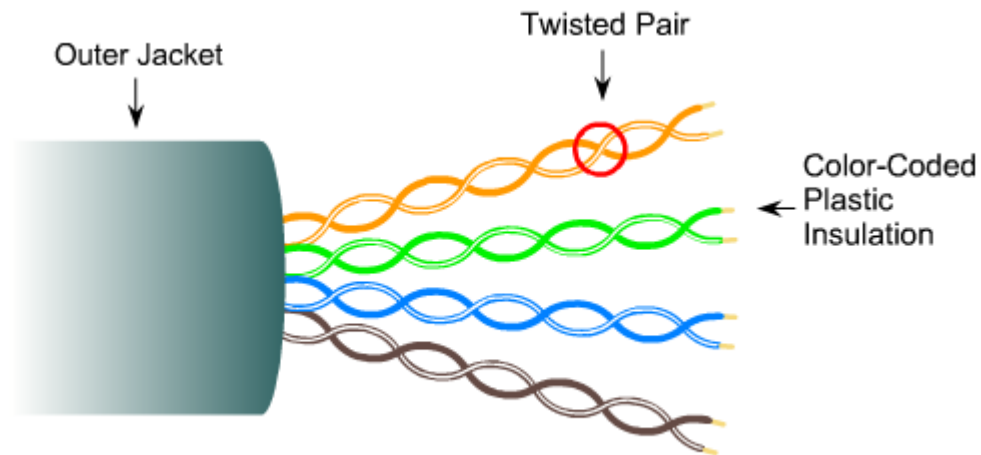
STP



Unshielded Twisted pair

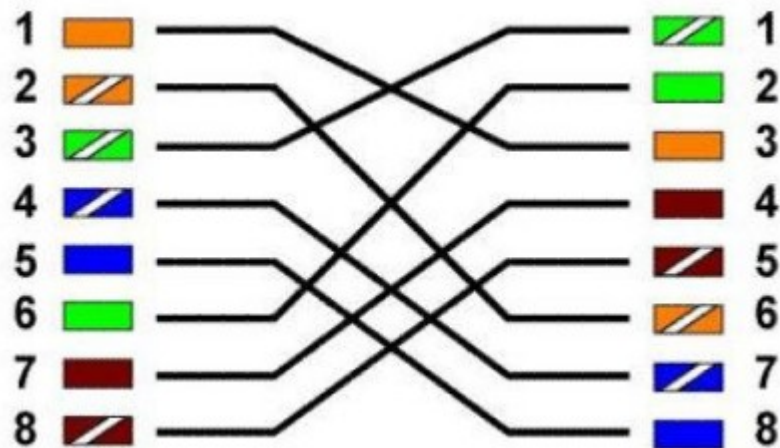
- ❑ Eight cables, Four pairs
- ❑ Each cable is covered with insulating material
- ❑ Each pair is twisted around each other for cancellation effect.
- ❑ Advantages include
- ❑ Speed 10 – 100 – 1000 Mbps (depend on category)
- ❑ Less expensive and easy to install.
- ❑ Maximum length 100 m
- ❑ Uses RJ-45 connector.
- ❑ Electric noise may occur.

Unshielded Twisted pair



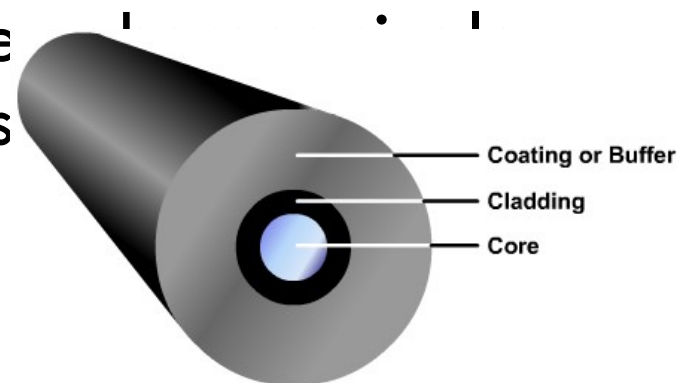
UTP cable

- Straight through cable (different devices)
- Crossover cable (similar devices)
- Rollover cable (RJ-45 to DB-9)



Optical Fiber

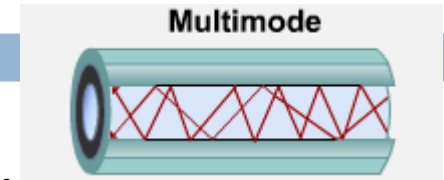
- Data or information is transmitted as light pulses.
- Carries more data for longer distances and much more speed as compare to other media.
- Requires more protection.
- There are two modes of optical fiber.
- Multimode
- Single mode
- Multimode used for short distance
mode is used for longer distances



Optical Fiber

- Optical fiber is not affected by outer noise.
- No crosstalk.
- Attenuation is caused by tight bends
- Bends causes cracks in the cladding and light rays are scattered.
- Scattering, absorption, dispersion, improper installation causes fiber losses.

Multimode optical fiber

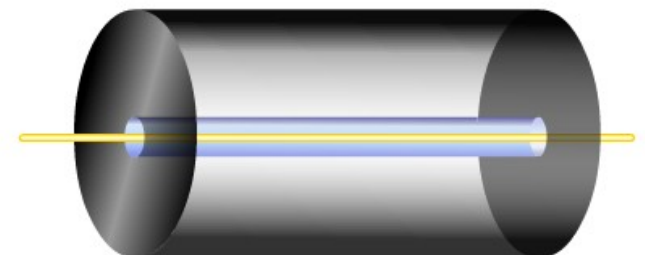


- Multimode operates at multiple beams.
- core in diameter is larger.
- Multimode has two forms:
 - Step index optical fiber
 - Graded index
- Two glass fibers are used for two way communication



Single mode Optical fiber

- Only allows one beam of light to travel
- Core is smaller in diameter.
- Light beam travels in the middle of the core.
- Single mode has higher data rates and greater speed.
- Single mode can carry data up to 3000m.



Unguided media

- Based on electromagnetic waves
- Do not use any physical conductor
- Signals are broadcast
- Electromagnetic spectrum
 - ▣ Radio waves & micro waves :3kHz to 300GHz
 - ▣ Infrared waves: 300GHz to 400GHz
- Ways in which signals travel from source to destination.
 - ▣ Ground propagation (low frequency signals)
 - ▣ Sky propagation (higher frequency signals, reflected back to earth)
 - ▣ Line of sight propagation (very high frequency signals, directed from antenna to antenna)

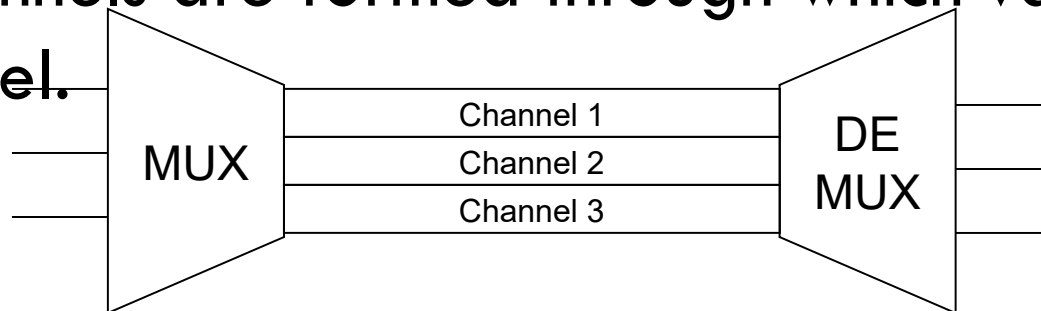
Multiplexer

- ❑ Make good use of available bandwidth.
- ❑ Simultaneous transmission of multiple signals across a single data link.
- ❑ n lines share the bandwidth of one link.
- ❑ Saves cost of multiple channels.
- ❑ We combine mux and De-mux into a single unit.
- ❑ Types of multiplexer
 - ❑ Frequency division
 - ❑ Time-division
 - ❑ Wavelength division



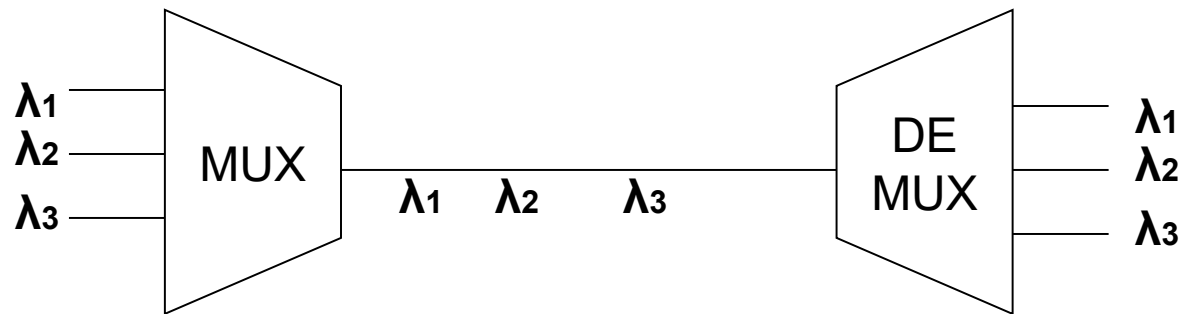
Frequency division

- When bandwidth (Hz) of link is greater than combined bandwidth of signals.
- Each sending device modulates signals at different carrier frequency.
- Modulated signals are combined into a single signal.
- Channels are formed through which various signals travel.



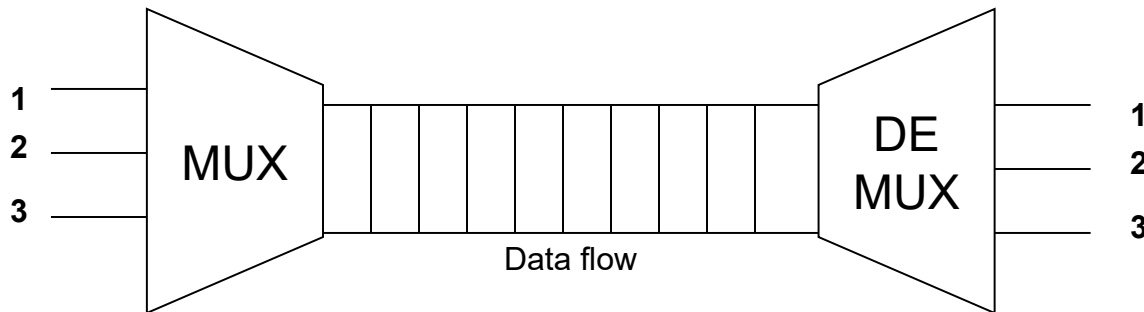
Wavelength-Division multiplexing

- Designed to use high data rates like optical fiber.
- Multiplexing allows to combine several lines into one.
- Same as FDM but operates optical signals instead of frequency signals.



Time division multiplexing

- Instead of sharing portion of bandwidth as in FDM, time is shared.
- Each connection occupies a portion of time in link.



Spread Spectrum

- We combine different sources to fit in larger bandwidth.
- But used in wireless applications.
- Wireless application uses air as medium for communication.
- Frequency of transmitted signal varies which results in higher bandwidth than required.
- So it spreads the original spectrum.
- conventional wireless systems remain at a fixed frequency. E.g. 101 MHz not goes up to 105MHz, location can be identified.
- Two types
 - Frequency hopping spread spectrum
 - ▣ Signal is modulated by set of frequencies to expand bandwidth.
 - Direct sequence spread spectrum
 - ▣ Each bit is assigned a code of n bits to increase the bandwidth.

IPv4 Addressing

- **Class A** addresses begin with **0xxx**, or **1 to 126** decimal.
- **Class B** addresses begin with **10xx**, or **128 to 191** decimal.
- **Class C** addresses begin with **110x**, or **192 to 223** decimal.
- **Class D** addresses begin with **1110**, or **224 to 239** decimal.
- **Class E** addresses begin with **1111**, or **240 to 254** decimal.

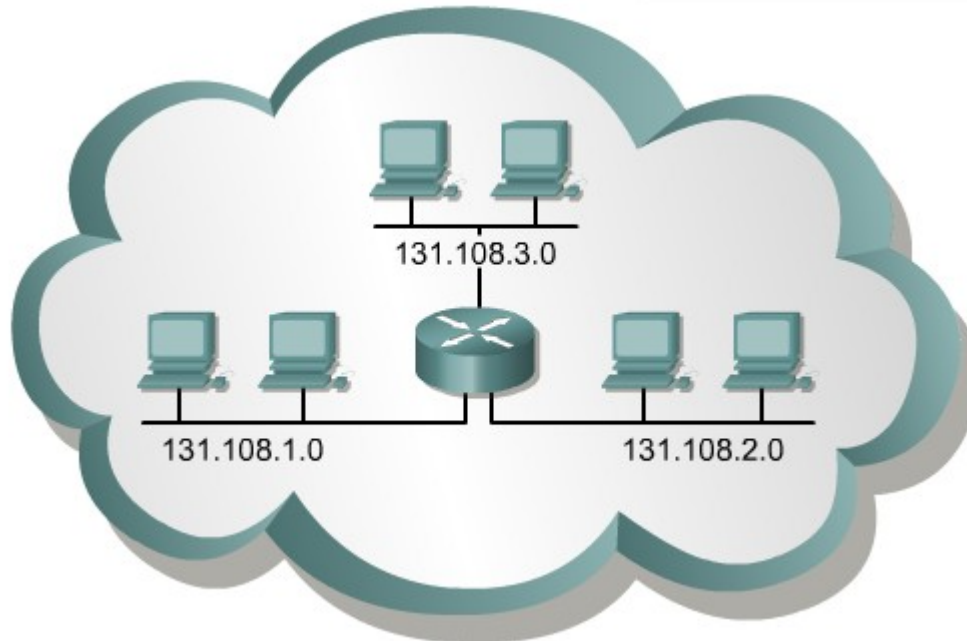
Parts of IP address belong to Network

- Class A -- NNNNNNNNN.nnnnnnnnn.nnnnnnnnn.nnnnnnnnn
- Class B --
NNNNNNNNNN.NNNNNNNNN.nnnnnnnnn.nnnnnnnnn
- Class C --
NNNNNNNNNN.NNNNNNNNN.NNNNNNNNN.nnnnnnnnn
- Each network IP has two parts
 - ▣ Network where system is connected
 - ▣ System itself

Address Class	Number of Networks	Number of Host per Network
A	126 *	16,777,216
B	16,384	65,535
C	2,097,152	254
D (Multicast)	N/A	N/A

Subnetting

- Dividing and identifying separate networks through LANs
- Prevents complete address exhaustion.
- Breaks



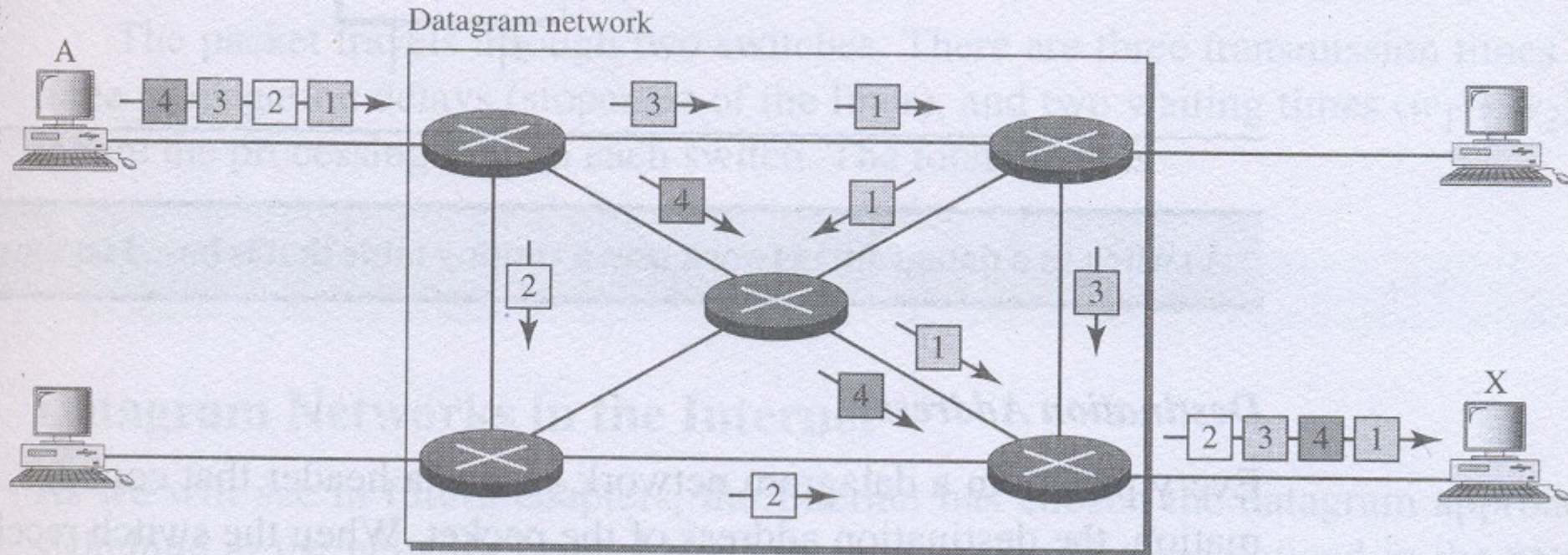
- $2^n - 2$ is the formula used to calculate total number of subnets and nodes.
- CIDR: Classless InterDomain Routing
- Example: we need 32 IP

# bits	Subnet Mask	CIDR	# Subnets	# Hosts	Nets * Hosts
2	255.255.255.192	/26	2	62	124
3	255.255.255.224	/27	6	30	180
4	255.255.255.240	/28	14	14	196
5	255.255.255.248	/29	30	6	180
6	255.255.255.252	/30	62	2	124

Datagram Networks

- Each message transmitted is converted into different packets of same sizes.
- Each packet is treated independently.
- Packets in this approach are referred to as *datagram* .
- Do not follow same path.
- Reach at destination in out of order.
- Datagram are connectionless.
- No setup or teardown phases.
- Routing table is used to send packets from source to destination.

Case study



- 
- Efficiency: better than circuit switching network.
 - Resources can be controlled, only used when transmitting packets.
 - Delay: datagram network has greater delay than circuit switching network.
 - Have to wait at each switch before transmission.

Virtual circuit networks

- Combination of circuit and packet switching networks.

Has following properties.

- Setup and teardown connection, like circuit switching, before data transfer.
- Resources are allocated during setup phase (circuit) or on demand (packet).
- Data is divided into datagram packets.
- But all packets follow same path.

- Has following processes.

- Setup
- Data transfer
- Acknowledgement
- teardown

Random Access method

- Each station is independent and can send data at any time.
- Has different protocols
 - ▣ ALOHA
 - ▣ CSMA/CD
 - ▣ CSMA/CA
- ALOHA
 - ▣ Developed earlier in 1970
 - ▣ Each station can send frame at any time.
 - ▣ There is only one channel
 - ▣ Collision possible.
 - ▣ ALOHA relies on acknowledgements
 - ▣ If ACK not received after time out period sender assumes frame destroyed it resends.
 - ▣ If all nodes resend at same time again collision possible.
 - ▣ So each station waits for random amount of time.